

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
DATA SCIENCE AND ANALYTICS
(7th SEMESTER, 4TH Year) LAB EXPERIMENT # 2

Lab # 2: To introduce Google Colab Notebook, and with NumPy Library.

Duration: 3 Hours

Objective: The objective of this lab session is to introduce students to the fundamental concepts of NumPy and library Python. Students will learn how to create and manipulate NumPy arrays.

Expected Outcomes: By the end of this lab session, students will be able to:

1. **NumPy:** Create and manipulate NumPy arrays, perform basic array operations, and apply functions such as reshaping, indexing, and mathematical operations.

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

1. Introduction

Google Colab is a free cloud-based platform developed by Google that allows users to write and execute Python code directly in a web browser. It comes with pre-installed libraries such as NumPy, Pandas, Matplotlib, TensorFlow, and PyTorch, making it ideal for Data Science, Machine Learning, and Big Data Analytics.

Unlike local development environments, Google Colab requires no software installation.

Understanding the Colab Interface

Getting Started with Google Colab

Opening Google Colab

1. Go to <https://colab.research.google.com/>
2. Sign in with your Google account.

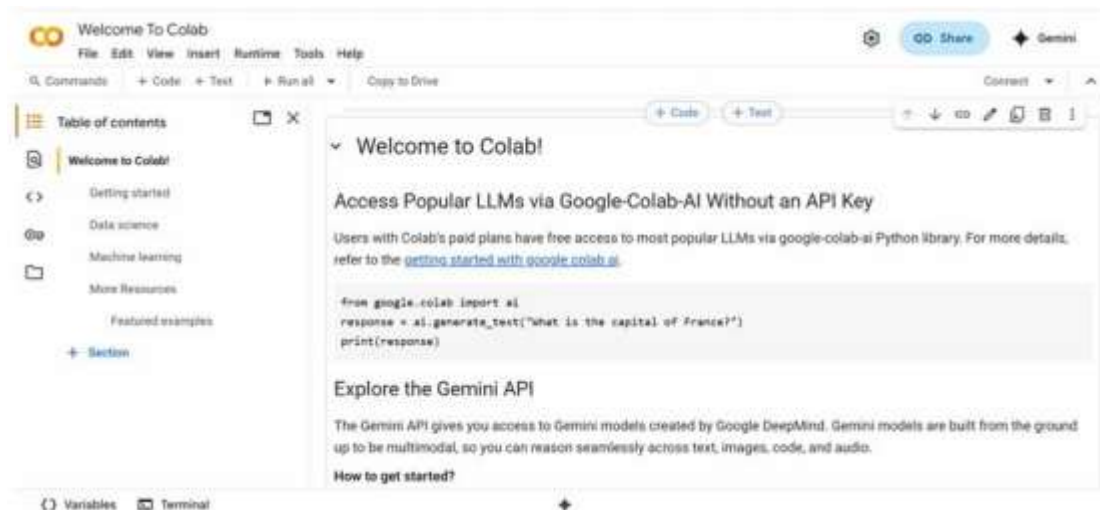
You'll see the Colab interface with options like:

- **New Notebook** – to create a blank notebook
- **Upload Notebook** – to open a notebook shared with you

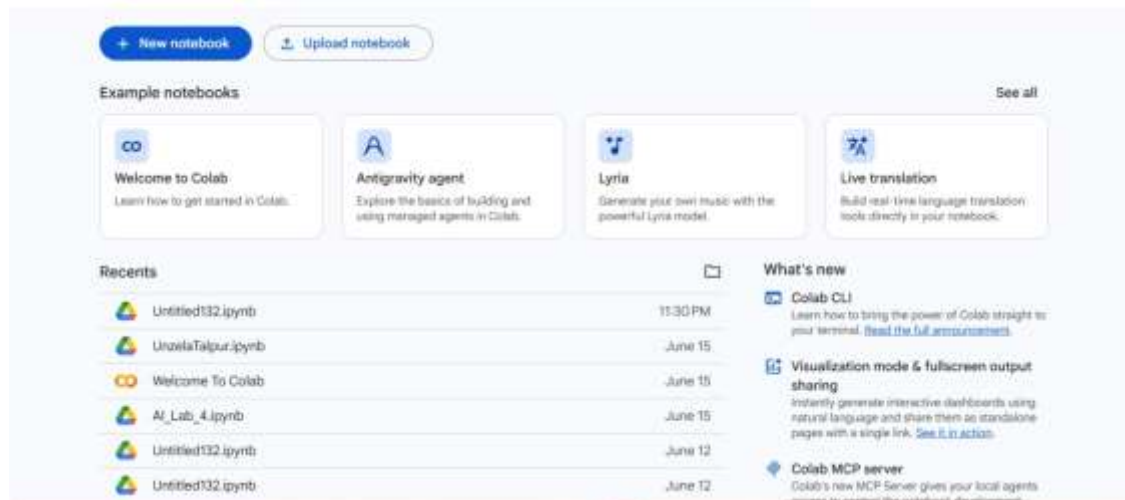
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- **Recent** – notebooks you have worked on before

Colab Interface



Or updated version

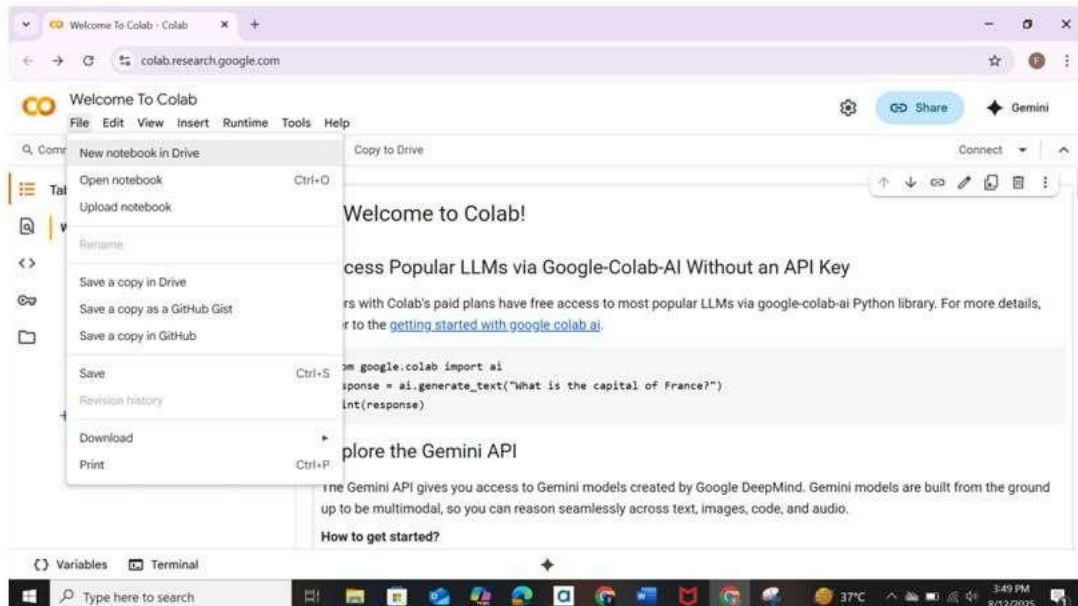


Creating a New Notebook

1. Click File → New Notebook or click New Notebook from the welcome screen.

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2. At the top left, click on Untitled0.ipynb and rename it to something meaningful 3. (e.g., First_Lab.ipynb).



Writing and Running Code

Colab notebooks have cells, you can type your code in them.

1. In a code cell, type:
 - `print ("Hello, Colab!")`
2. To run the cell:
 - Click the Play ► button on the left or Press Shift + Enter on your keyboard.

The output will appear right below the cell.



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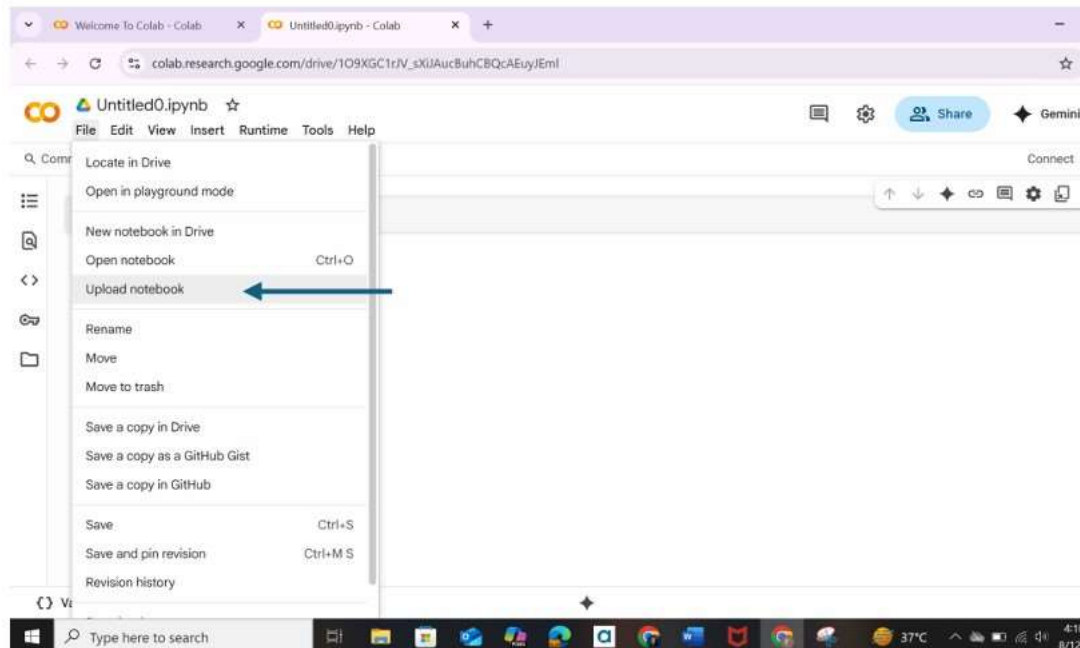
Uploading a Notebook

1. Download the notebook file (.ipynb) from my shared link.
2. In Colab, click File → Upload Notebook.
3. Choose the file from your computer and open it.

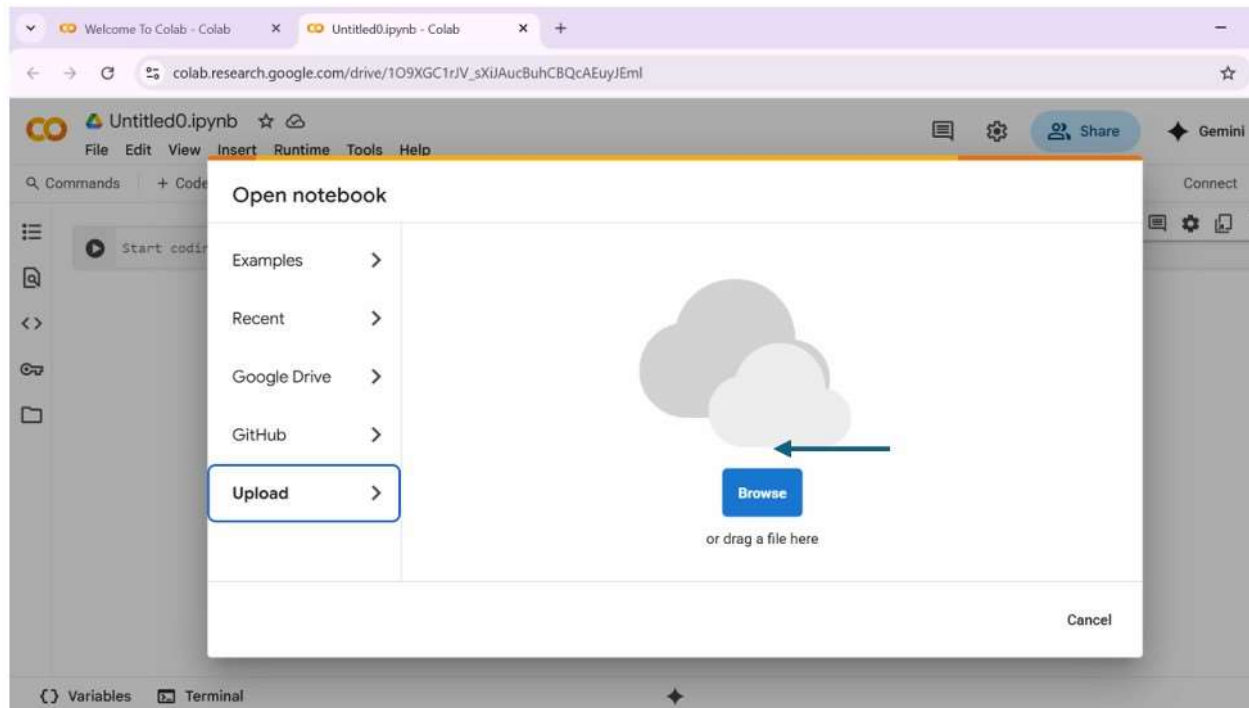
Saving Your Work

Colab automatically saves your work in your Google Drive.

You can also download it to your computer via File → Download → Download .ipynb



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Renaming a File in Colab

Method 1: Rename Notebook from the Top Bar

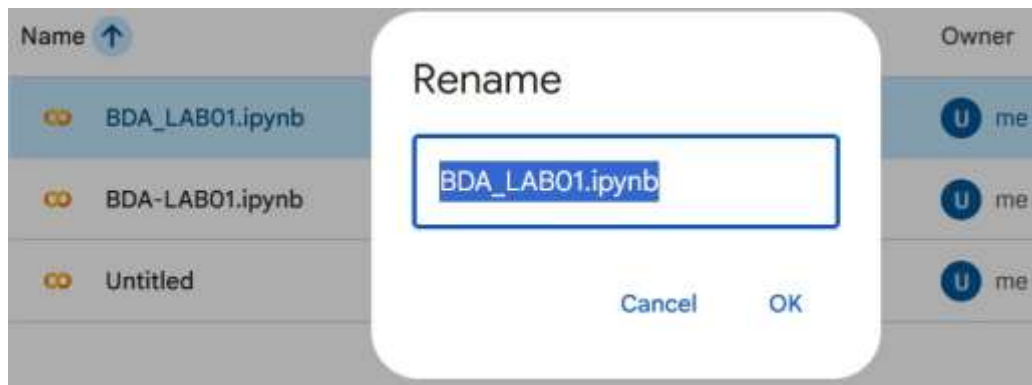
1. At the top-left of Colab, you'll see the notebook name (by default something like Untitled0.ipynb).
2. Click on the name.
3. A popup will appear → type the new name (e.g., Lab1_Intro.ipynb).
4. Press Enter to save.

Method 2: Rename from Google Drive

Since Colab notebooks are saved to your Google Drive:

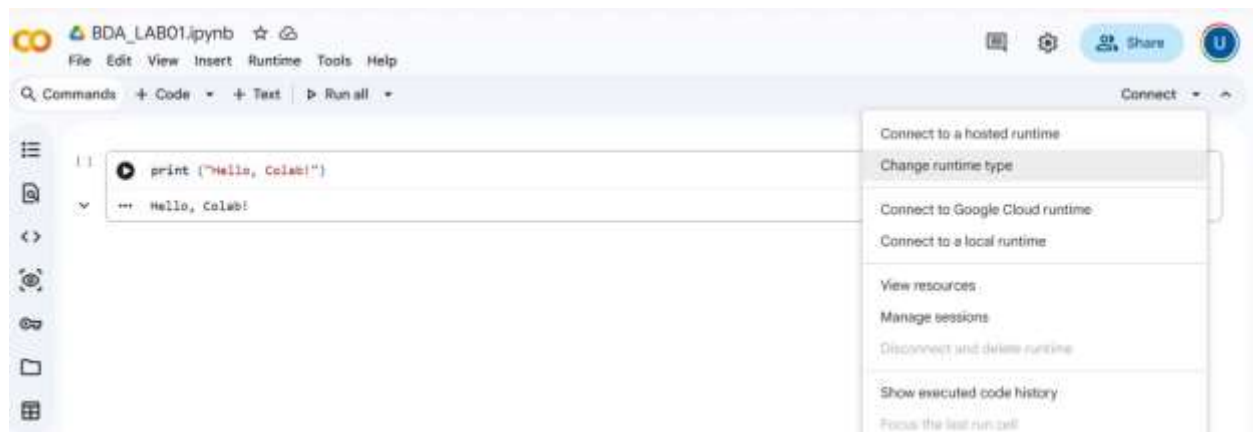
1. Go to Google Drive.
2. Locate your notebook file.
3. Right-click → Rename.
4. Enter the new name and save.

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Change Runtime in Google Colab

1. Open your Colab notebook.
2. Go to the top menu bar → click on Runtime.
3. From the dropdown, select Change runtime type.
4. A popup window will appear:
 - Runtime type: Choose Python 3 (default) or another if available.
 - Hardware accelerator: Select one of the following:
 - ✦ None → Runs on CPU (default).
 - ✦ GPU → GPU (Graphics Processing Unit) Use this if your code requires graphics processing (like neural networks).
 - ✦ TPU → Tensor Processing Unit (for advanced deep learning tasks). Click Save



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Change runtime type

Runtime type

Python 3

Hardware accelerator ?

☒ CPU ☐ T4 GPU ☐ H100 GPU ☐ G4 GPU

☐ A100 GPU ☐ L4 GPU ☐ v5e-1 TPU

☐ v6e-1 TPU

Want access to premium GPUs? [Purchase additional compute units](#)

Runtime version ?

Latest (recommended)

Cancel Save

Benefits of Using Google Colab

- Free to use; no installation or setup needed.
- Runs in the cloud can work even on low-spec computers.
- Accessible anywhere just log in with your Google account.
- Pre-installed libraries with popular Python packages like NumPy, Pandas, Matplotlib, and TensorFlow are ready to use.
- Easy sharing and collaboration just like Google Docs, you can share notebooks with others in real time.
- Supports interactive coding, run code, see output immediately, and visualize results in the same window.

GPU/TPU acceleration speed up heavy computations without needing expensive hardware **NumPy:**
Numerical Operations

What is NumPy?

NumPy (Numerical Python) is a Python library used for working with numbers and performing mathematical calculations quickly.

The main feature of NumPy is the **array**.

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A NumPy array is a collection of numbers stored in an organized way, similar to a list, but it is **faster and uses less memory**.

NumPy array:

```
import numpy as np

numbers = np.array([1,2,3,4,5])
```

Why is NumPy Faster than Python Lists?

A normal Python list can store different types of data:

[10, "Ali", 3.5]

But NumPy arrays store only one type of data:

[10,20,30,40]

Because all elements have the same type, the computer can process them faster.

What is a Vectorized Operation?

A **vectorized operation** means performing an operation on **an entire array at once** instead of using a loop to process each element one by one.

Normally in Python, we use loops:

A screenshot of a Jupyter Notebook interface. The code cell contains the following Python code:

```
[8] ✓ Os ▶ a = [1,2,3]
b = [4,5,6]

result = []

for i in range(3):
    result.append(a[i]+b[i])
print(result)
```

The output of the cell is shown below the code:

```
▼ ... [5, 7, 9]
```

With NumPy:

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```
[10] import numpy as np

a = np.array([1,2,3])
b = np.array([4,5,6])

result = a + b
print(result)

... [5 7 9]
```

NumPy performs the operation on the complete array at once.

This is called **vectorization**.

Uses of NumPy in Data Science and Analytics

- Handling Numerical Data
- Fast Mathematical Computations
- Statistical Analysis

```
[2] #Check if NumPy and Pandas are Installed

import numpy as np
import pandas as pd

print('NumPy and Pandas are successfully imported!')

NumPy and Pandas are successfully imported!
```

If not installed, then install

```
#Installing NumPy (if required)
!pip install numpy

... Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
```

You can install both

```
!pip install numpy pandas

... Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

https://colab.research.google.com/drive/1R5GaQjcXw9H-XWQea_vl9cLvWSW_eNca?usp=sharing

access this notebook for your practice and understanding

EXERCISE

Instructions:

Prepared by Engr: Unzela Talpur

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1. Create your own **Google Colab Notebook**.
2. Rename the notebook using the format: **Your_Name_Roll_Number**

Example:

Arsalan_23BSCYS016

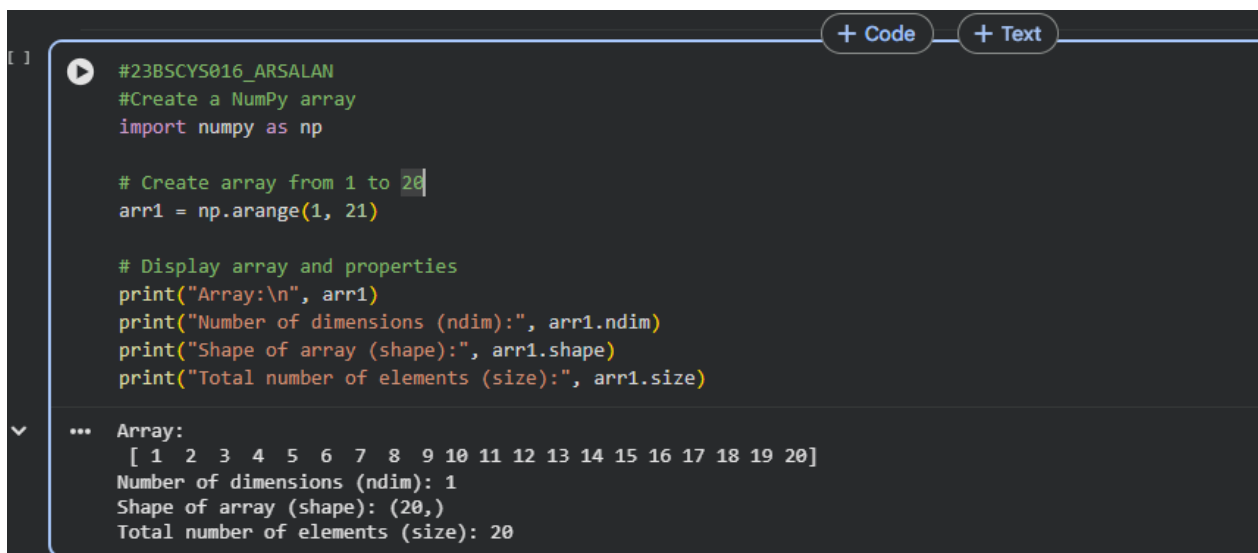
3. Solve all the given questions in your notebook.
4. Share the notebook with me.
5. Make sure the sharing permission is set to **Viewer only** (I should not have editing access).

Part 1: NumPy Exercises

Q1. Create a NumPy array

Create a NumPy array containing numbers from 1 to 20.

- Display the array.
- Find and display:
 - ✦ Number of dimensions (ndim)
 - ✦ Shape of the array (shape)
 - ✦ Total number of elements (size)



The screenshot shows a Google Colab notebook interface. At the top right, there are two buttons: '+ Code' and '+ Text'. Below them is a code editor with a dark background and light-colored text. The code in the editor is as follows:

```
#23BSCYS016_ARSALAN
#Create a NumPy array
import numpy as np

# Create array from 1 to 20
arr1 = np.arange(1, 21)

# Display array and properties
print("Array:\n", arr1)
print("Number of dimensions (ndim):", arr1.ndim)
print("Shape of array (shape):", arr1.shape)
print("Total number of elements (size):", arr1.size)
```

Below the code editor, the output of the code is displayed. It starts with '...' followed by the array elements: [1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20]. Below the array, the properties are listed: 'Number of dimensions (ndim): 1', 'Shape of array (shape): (20,)', and 'Total number of elements (size): 20'.

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Q2. Array Reshaping

Create a NumPy array: [1,2,3,4,5,6,7,8,9,10,11,12]

Perform the following:

- Reshape it into a **3 × 4 matrix**.
- Display the reshaped array.
- Convert it into a 1D array using `flatten()`.

```
#Array Reshaping
import numpy as np

# Create array from 1 to 12
arr2 = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12])

# Reshape into 3 x 4 matrix
reshaped_arr = arr2.reshape(3, 4)
print("3x4 Matrix:\n", reshaped_arr)

# Convert back to 1D array using flatten()
flattened_arr = reshaped_arr.flatten()
print("Flattened 1D Array:\n", flattened_arr)
```

```
... 3x4 Matrix:
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
Flattened 1D Array:
[ 1  2  3  4  5  6  7  8  9 10 11 12]
```

Q3. Mathematical Operations

Create two NumPy arrays:

A = [10,20,30,40,50]

B = [5,10,15,20,25]

Perform: Addition, Subtraction, Multiplication, Division

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```
#Mathemmmatical Operations
import numpy as np

# Define arrays
A = np.array([10, 20, 30, 40, 50])
B = np.array([5, 10, 15, 20, 25])

# Perform operations
print("Addition (A + B):", A + B)
print("Subtraction (A - B):", A - B)
print("Multiplication (A * B):", A * B)
print("Division (A / B):", A / B)
```

```
... Addition (A + B): [15 30 45 60 75]
Subtraction (A - B): [ 5 10 15 20 25]
Multiplication (A * B): [ 50 200 450 800 1250]
Division (A / B): [2. 2. 2. 2. 2.]
```

Q4. Find Statistics

Create an array of student marks:

[78,85,90,66,95,88,72]

Find: Maximum marks, Minimum marks, Average marks, Median marks

```
#Find Statistics
import numpy as np

# Array of student marks
marks = np.array([78, 85, 90, 66, 95, 88, 72])

# Calculate statistical values
print("Maximum Marks:", np.max(marks))
print("Minimum Marks:", np.min(marks))
print("Average Marks:", np.mean(marks))
print("Median Marks:", np.median(marks))
```

```
... Maximum Marks: 95
Minimum Marks: 66
Average Marks: 82.0
Median Marks: 85.0
```

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Colab Link:

<https://colab.research.google.com/drive/1BTrkfdN37BPzfLd8CP6sapQCBqeQOHwH?usp=sharing>

Submission Checklist

Before submitting, make sure:

- ☐ Notebook name contains your **Name + Roll Number**
- ☐ All questions are solved
- ☐ Code output is visible
- ☐ Notebook is shared as **Viewer only**
- ☐ Link is submitted